# Waves at boundaries

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Describe reflection and transmission at a boundary.

**SUCCESS CRITERIA - I can...**

1. Identify incident, reflected and transmitted waves.
2. Distinguish fixed-end from free-end pulse reflection.
3. Track energy without assuming equal amplitudes.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “boundary” and “reflection” to explain your model or result. Sentence starters are optional.

**boundary**: the place where a medium or constraint changes

**reflection**: a wave returning from a boundary

**transmission**: a wave continuing through or beyond a boundary

## 01 Warm-up: bring back an earlier idea

Echoes return at 0.003 s and 0.005 s with the same wave speed.
**Compare** ranges. Does delay alone determine mass?

**Word help:** wave = a disturbance that can carry energy and information

**Sentence starter:** Both ... . They differ in ... because ... .

**Marzano check - circle:** ① ② ③

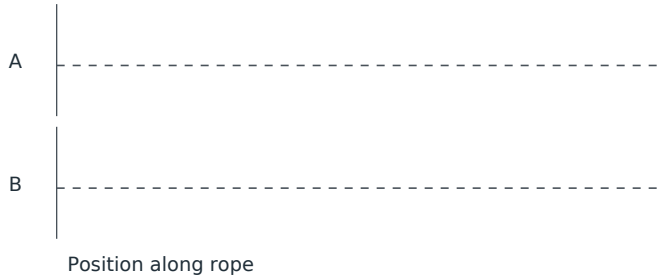
## 02 Worked example: explain the model

**Reference:** $f_{reflected} = f_{incident}$.
At a stationary boundary. A displacement pulse returns inverted from an ideal fixed end and upright from an ideal free end.

**Worked reasoning:** At a fixed end, an upward rope pulse reflects downward. At an ideal free end it reflects without inversion. These are different boundary conditions.

**Your turn: Draw** incident and reflected pulse snapshots for both ends; show propagation directions.

**Sentence starter:** The ... in my model represents ... . Its label shows ... .




**Marzano check - circle:** ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 03 Read and interpret the model

An upward rope pulse reaches an ideal fixed end.
**Sketch** the reflected pulse and its travel direction.

**Word help:** ideal = using stated simplifying conditions

**Sentence starter:** The ... in my model represents ... . Its label shows ... .



Sketch and label your response.

**Marzano check - circle:** 1 2 3

## 04 Apply the physics

A pulse carries 0.80 J to a boundary. Of this, 0.20 J is reflected and 0.10 J is absorbed. Complete the energy-flow model.
**Find** transmitted energy and the reflected, absorbed and transmitted fractions.

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

**Marzano check - circle:** 1 2 3

## 05 Explain your reasoning

Does a smaller transmitted amplitude prove frequency changed?
**Explain** what usually fixes the frequency at a stationary boundary.

**Sentence starter:** Both ... . They differ in ... because ... .

**Marzano check - circle:** 1 2 3

## 06 Review: connect an earlier idea

A 0.4 kg vapor releases 120000 J while condensing at constant temperature.
**Find** latent heat and signed Q for the sample.
**Label** temperatures, samples and relevant energy transfers.

**Word help:** latent heat = energy per mass needed for a specified phase change

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

**Marzano check - circle:** 1 2 3

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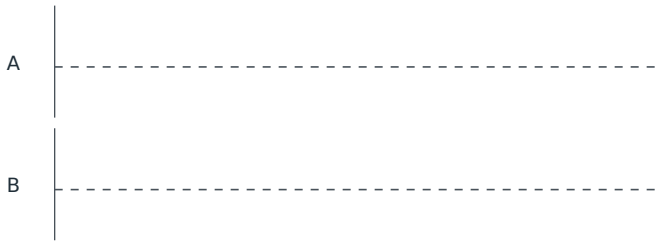
Continue down each column.

Sentence starters are optional. You may plan in any language.

## 07 Exit check

At a stationary boundary frequency stays 5 Hz while wave speed changes from 10 to 20 m/s.
**Find** wavelength on each side.

**Sentence starter:** I found ... [unit]. This value means ... in this situation.




**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the model

A pulse reflects from a fixed end and meets an equal incident pulse. At an instant their displacements are +2 and -2 cm at one point.
**Find** total displacement and explain why zero there does not mean all energy vanished.

**Sentence starter:** Under the assumption ..., the evidence ... supports ... because ... .



Sketch and label your response.

**Marzano check - circle:** 1 2 3